

Review of R

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We've noticed that some of you haven't yet taken AMS 597 Statistical Computing, which means a refresher on R and routine data analysis techniques will be necessary. This isn't a problem—we anticipated it. Over the next two weeks, we'll review the essential elements of R for statistical analysis during class sessions.

For those already comfortable with R, please begin exploring machine learning by reading textbooks. A good starting point is:

"An Introduction to Statistical Learning with Applications in R" (2017) by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani (Springer).

Our initial focus will be on:

- Basic R commands
- Fundamental statistical analysis using R

In addition, please start familiarizing yourselves with one another and form teams, since the team project accounts for 20% of your grade. We plan to form **eight teams**, each comprising about **ten members** from varied backgrounds (different programs/tracks). The presentations (PowerPoint) will commence in mid-April. Each team should choose a leader and submit a roster with each member's name, ID, and track information.

If needed, teams can organize themselves, or I can help randomly assign teams. For guidance on R—including topics like data cleaning or linear regression—please refer to this manual:

<https://www.scribbr.com/category/statistics/>

Also, review the AMS 572 (Data Analysis) lecture notes (by Prof. Zhu) for foundational statistical methods here:

http://www.ams.sunysb.edu/~zhu/wei/Teaching_Fall2013.html.

After covering these basics, we will move on to other statistical learning methods. For further reading and resources, visit:

<https://tdg5.github.io/stats-learning-notes/>.